

5-Mark Questions & Answers -- DBMS Module 4: Relational Database Design

Q1. Explain 1NF, 2NF, 3NF, and BCNF with concrete table examples.

1. 1NF (Atomic Values): Eliminates multi-valued cells or comma-separated lists.

2. 2NF

Q2. Explain the 3NF synthesis algorithm for lossless-join and dependency-preserving decomposition.

1. Compute Minimal Cover F_c for functional dependency set F .
2. For each FD $(X \rightarrow Y)$ in F_c :
Create a new relation schema $R_i = X \cup Y$.
3. If no schema R_i contains a Candidate Key of original relation R :
Create an additional schema containing any Candidate Key of R .
4. Remove redundant sub-schemas (if R_i is a subset of R_j , delete R_i).

* *Guarantee:* Resulting schemas are in 3NF, preserve all functional dependencies, and guarantee lossless join.

Q3. Explain 4NF and decomposition using multivalued attributes with an example.

- * 4NF Condition: A BCNF relation is in 4NF if for every non-trivial MVD $X \twoheadrightarrow Y$, X is a super key.

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Q4. Describe NoSQL Data Models (Key-Value, Document, Column-Family, Graph) with use cases and CAP theorem.

- * CAP Theorem: Distributed systems can achieve at most 2 out of 3: Consistency, Availability, Partition Tolerance.

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